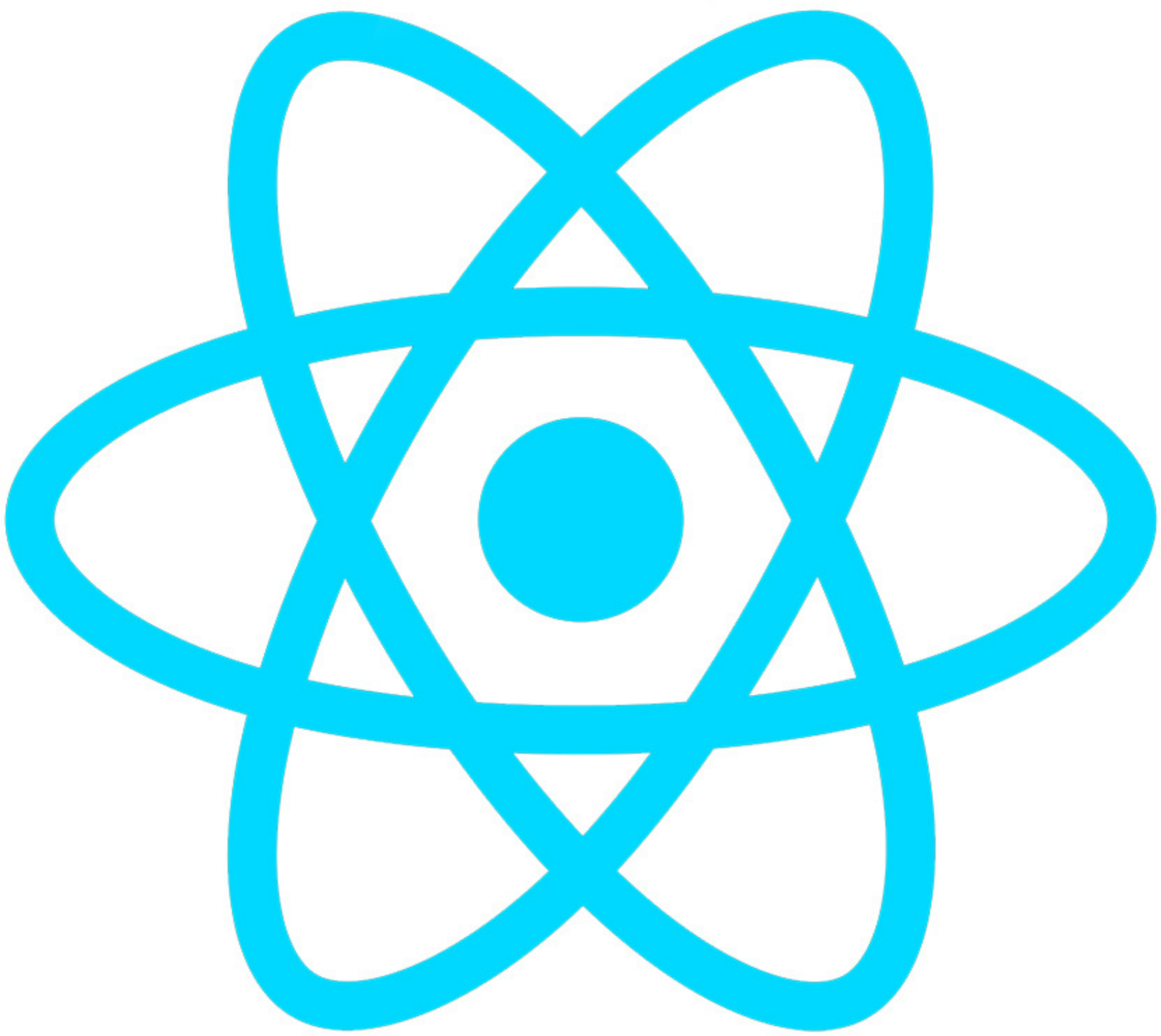
TOP 50

REACT

Interview Questions & Answers



Q 1. What is React?

Ans: React is an open-source JavaScript library for building user interfaces or UI components, developed by Facebook.

Q 2. Explain JSX

Ans: JSX (JavaScript XML) is a syntax extension for JavaScript recommended by React for describing what the UI should look like.

Q 3. What is the virtual DOM?

Ans: The virtual DOM is a lightweight copy of the actual DOM in memory. React uses it to improve performance by updating only the changed parts of the actual DOM.

Q 4. What is the significance of keys in React?

Ans: Keys are used to uniquely identify and differentiate between components in React. They help React identify which items have changed, added, or removed.

Q 5. What are state and props in React?

Ans: State is an internal data store that belongs to a specific component, and it can be changed over time. Props are properties passed to a component from its parent, and they are immutable.

Q 6. What is the difference between state and props?

Ans: State is internal to a component and can be changed over time, while props are external and passed to a component.

Q 7. Explain the concept of lifting state up.

Ans: Lifting state up is a pattern where the state of a child component is moved to its parent component, allowing multiple child components to share the same state.

Q 8. What is the purpose of setState in React?

Ans: setState is used to update the state of a component and trigger a re-render.

Q 9. What is React Router?

Ans: React Router is a library that enables navigation among views in a React application, allowing for the development of single-page applications.

Q 10. Explain the useEffect hook.

Ans: The useEffect hook in React is used for side effects in functional components, such as data fetching, subscriptions, or manually changing the DOM.

Q 11. What are controlled components in React?

Ans: Controlled components are components where the form data is controlled by React state. The input elements receive their current value from the state and have their value updated through a callback function.

Q 12. What is Redux, and why is it used?

Ans: Redux is a state management library for JavaScript applications, commonly used with React. It helps manage the state of an application in a predictable way.

Q 13. Explain the concept of higher-order components (HOC).

Ans: Higher-order components are functions that take a component and return a new component with additional features or props.

Q 14. What is the purpose of the useReducer hook?

Ans: The useReducer hook is used for managing complex state logic in React applications. It is an alternative to useState when state transitions are more complex.

Q 15. What is the significance of the key attribute in React lists?

Ans: The key attribute is used to uniquely identify elements in a list. It helps React efficiently update the DOM when the list changes.

Q 16. What is the difference between class components and functional components?

Ans: class components use ES6 classes and have additional features like state and lifecycle methods, while functional components are simpler and are often used with hooks.

Q 17. Explain the concept of refs in React.

Ans: Refs are used to access the DOM directly or to reference a React element. They provide a way to interact with the underlying DOM nodes in React.

Q 18. What are React hooks?

Ans: React hooks are functions that allow functional components to use state, lifecycle methods, and other React features.

Q 19. Explain the purpose of the useContext hook.

Ans: The useContext hook is used to access the value of a React context within a functional component.

Q 20. What is the significance of the dangerouslySetInnerHTML property in React?

Ans: dangerouslySetInnerHTML is used to inject HTML directly into a component, but it should be used with caution to avoid cross-site scripting (XSS) vulnerabilities.

Q 21. What is the purpose of the componentDidMount lifecycle method?

Ans: componentDidMount is invoked immediately after a component is mounted, making it suitable for initial AJAX requests or setting up subscriptions.

Q 22. What is the React developer tool?

Ans: The React Developer Tools is a browser extension that allows developers to inspect and debug React component hierarchies in the Chrome and Firefox browsers.

Q 23. Explain the concept of context in React.

Ans: Context provides a way to pass data through the component tree without having to pass props manually at every level. It is often used to share values like themes or authentication status.

Q 24. What are the advantages of using React?

Ans: React offers a virtual DOM for improved performance, a component-based architecture for modular development, and a strong community support, among other advantages.

Q 25. How does React handle prop drilling, and how can it be avoided?

Ans: Prop drilling occurs when props are passed down through multiple levels of components. It can be avoided by using context or state management libraries like Redux.

Q 26. What is the purpose of the shouldComponentUpdate method?

Ans: shouldComponentUpdate is a lifecycle method that determines if a component should re-render. Developers can use it to optimize performance by preventing unnecessary renders.

Q 27. Explain the significance of React Fragments.

Ans: React Fragments allow developers to group multiple elements without adding an extra node to the DOM, helping to keep the structure clean.

Q 28. What is the significance of the key prop in React Router?

Ans: The key prop in React Router is used to force the remounting of a component when the key changes, ensuring that the component is fully reinitialized.

Q 29. What is the purpose of the forwardRef function in React?

Ans: forwardRef is used to forward refs through components, allowing parent components to interact with the child's DOM node.

Q 30. Explain the concept of error boundaries in React.

Ans: Error boundaries are components that catch JavaScript errors anywhere in their child component tree and log those errors, display a fallback UI, or take other actions.

Q 31. What is the significance of the memo function in React?

Ans: memo is a higher-order component that memoizes the rendering of a functional component, preventing unnecessary re-renders if the props have not changed.

Q 32. How does React handle forms?

Ans: React handles forms by using controlled components, where form data is controlled by the React state.

Q 33. Explain the purpose of the useMemo hook.

Ans: The useMemo hook is used to memoize the result of a function, preventing unnecessary calculations and improving performance.

Q 34. What is the significance of the useCallback hook?

Ans: useCallback is used to memoize callback functions, preventing them from being recreated on every render.

Q 35. What are React portals?

Ans: React portals provide a way to render children into a DOM node that exists outside the parent component's hierarchy.

Q 36. Explain the concept of suspense in React.

Ans: Suspense is a feature in React that allows components to "wait" for something before rendering, such as data fetching or code splitting.

Q 37. What is the purpose of the useEffect cleanup function?

Ans: The cleanup function in useEffect is used to perform cleanup tasks, such as unsubscribing from subscriptions or clearing intervals, when a component is unmounted.

Q 38. How does React handle routing?

Ans: React can handle routing using the React Router library, which provides a way to navigate between different views or pages in a React application.

Q 39. What is the purpose of the useLayoutEffect hook?

Ans: useLayoutEffect is similar to useEffect, but it fires synchronously after all DOM mutations. It is often used for measuring and synchronizing layout.

Q 40. Explain the concept of lazy loading in React.

Ans: Lazy loading is a technique where components or modules are loaded only when they are actually needed, improving initial load times.

Q 41. What is the significance of the React.memo function?

Ans: React.memo is a higher-order component that memoizes the rendering of a functional component, preventing unnecessary re-renders if the props have not changed.

Q 42. How does React handle code splitting?

Ans: React supports code splitting, allowing developers to split their code into smaller chunks that are loaded on demand, improving performance by reducing the initial bundle size.

Q 43. What is the purpose of the useImperativeHandle hook?

Ans: useImperativeHandle is used to customize the instance value that is exposed when using React.forwardRef.

Q 44. Explain the concept of the useDebugValue hook.

Ans: useDebugValue is used to display a label for custom hooks in React DevTools.

Q 45. What is the purpose of the useState hook?

Ans: The **useState** hook is used to add state to functional components in React.

Q 46. Explain the significance of the SuspenseList component in React.

Ans: SuspenseList is a component that allows developers to coordinate the loading of multiple components in a way that provides a better user experience.

Q 47. What is the significance of the react-scripts package in a React application?

Ans: react-scripts is a set of scripts and configurations used by Create React App to set up and manage a React project without the need for complex configuration.

Q 48. Explain the concept of error boundaries in React.

Ans: Error boundaries are components that catch JavaScript errors anywhere in their child component tree and log those errors, display a fallback UI, or take other actions.

Q 49. What is the purpose of the useReducer hook?

Ans: useReducer is a hook in React used for state management in functional components. It is particularly useful when the state logic is complex and involves multiple sub-values or when the next state depends on the previous state.

Q 50. How does React handle forms?

Ans: React handles forms using controlled components, where the form elements are controlled by React state. This allows React to be the single source of truth for the form data.